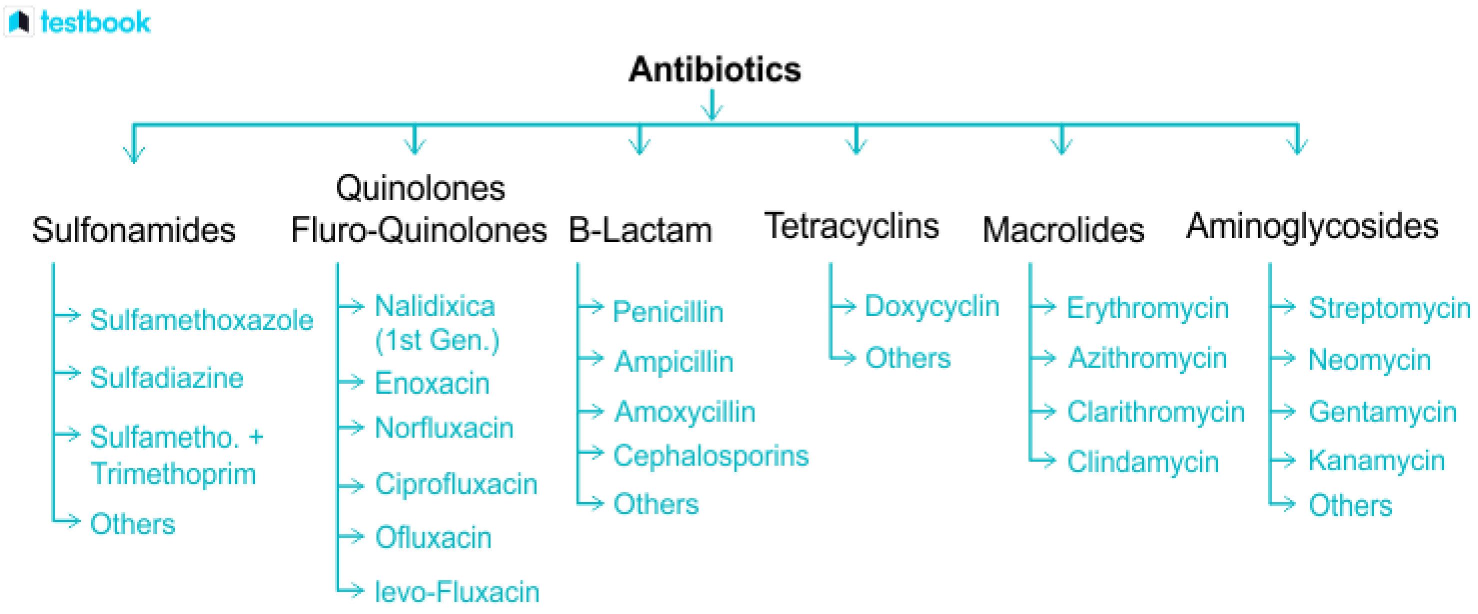
- Antimicrobials should only be given when necessary and after antimicrobials susceptibility test whenever possible.
- The pharmacokinetics of the drug should be taken into consideration e.g. the state of hepatic and renal functions of the patient.
- In serious infection it is better to start with a parentral loading of a bactericidal agent to avoid emergence of resistant strains by giving adequate dosage for sufficient duration and adapting proper combination regimens.
- Antimicrobials should be continued for 3 days after apparent cure is achieved to avoid relapse.


testbook
Antibiotics
Sulfonamides
Quinolones
Fluro-Quinolones
B-Lactam
Tetracyclins
Macrolides
Aminoglycosides
Sulfamethoxazole
Sulfadiazine
Sulfametho. +
Trimethoprim
Others
Nalidixica
(1st Gen.)
Enoxacin
Norfluxacin
Ciprofluxacin
Ofluxacin
levo-Fluxacin
Penicillin
Ampicillin
Amoxycillin
Cephalosporins
Others
Doxycyclin
Others
Erythromycin
Azithromycin
Clarithromycin
Clindamycin
Streptomycin
Neomycin
Gentamycin
Kanamycin
Others

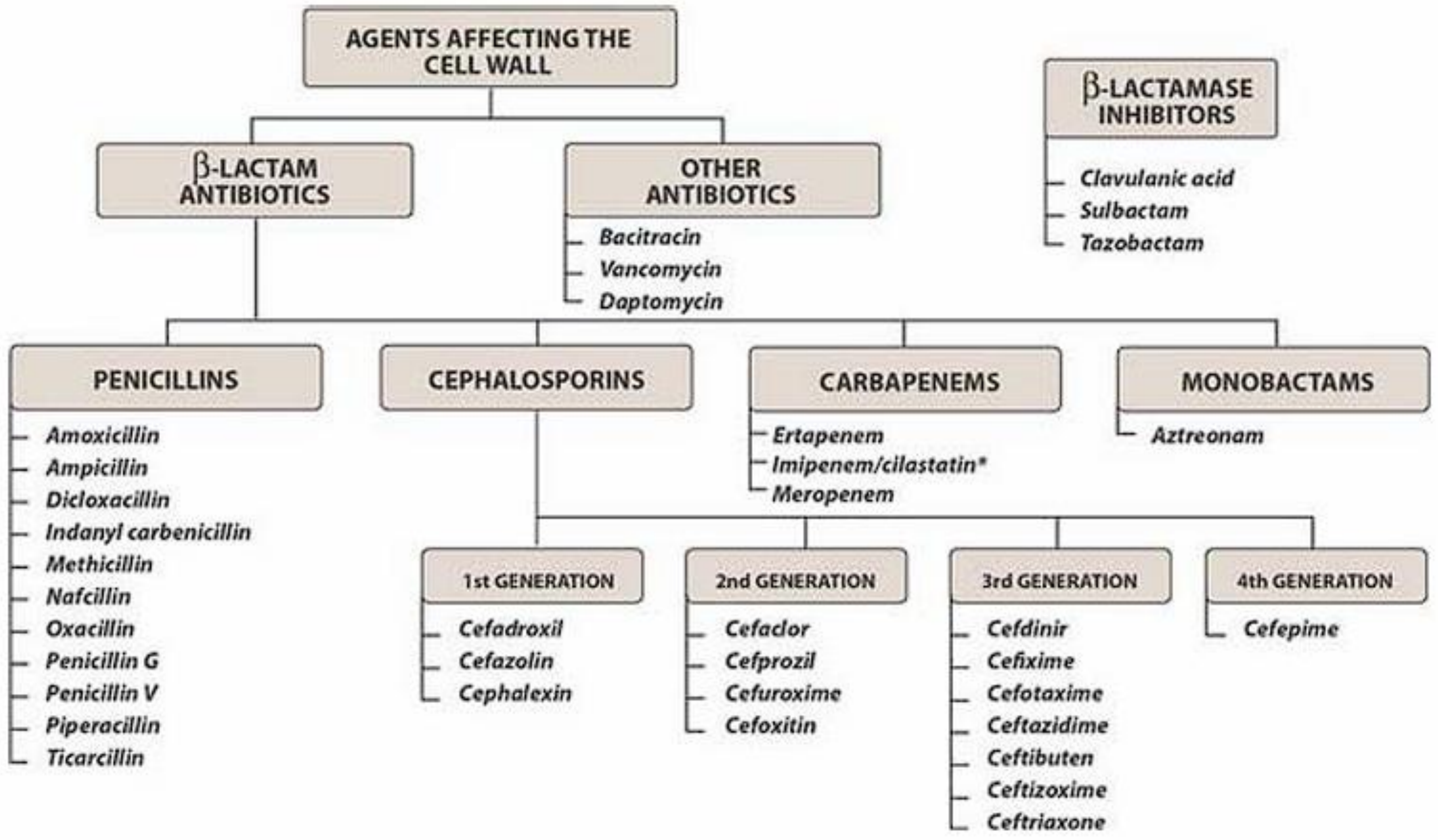


Figure 31.1 Summary of antimicrobial agents affecting cell wall synthesis. **Cilastatin* is not an antibiotic but a peptidase inhibitor that protects *imipenem* from degradation.

Pencillin Mechanism of action

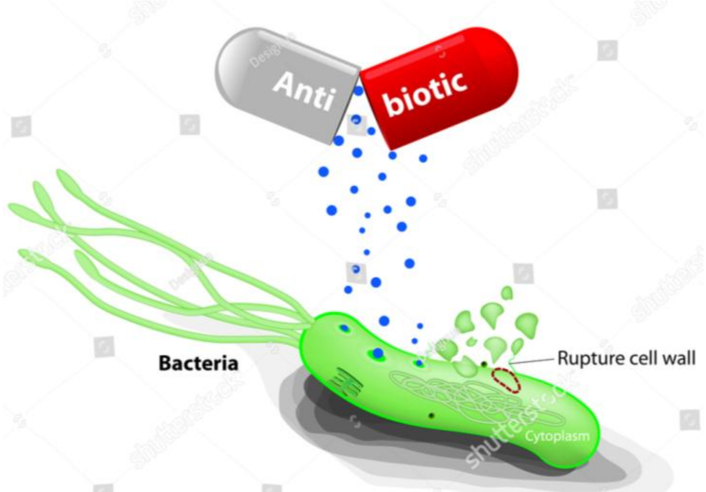
Pencillin+probenecid

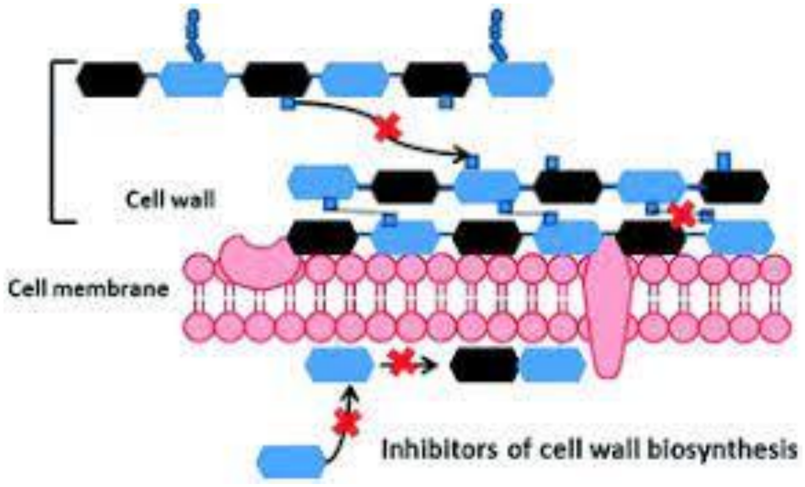
A. CELL WALL INHIBITORS

β-LACTAM ANTIBIOTICS

They share a β-lactam ring in their molecular structure which is required for antibacterial activity.

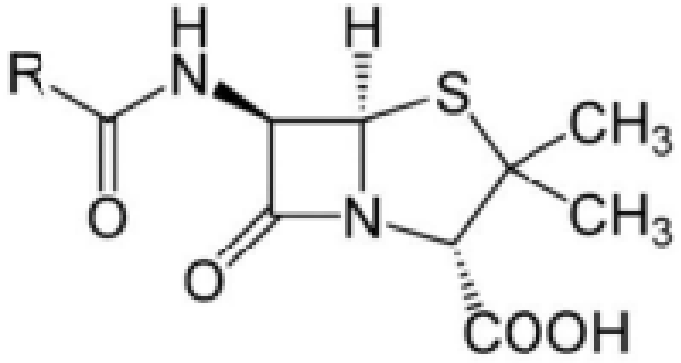
- **Penicillin**
- **Cephalosporins**
- **Monobactams**
- **Carbapenem**





.

Penicillins



Penicillin core structure, where "R" is the variable group

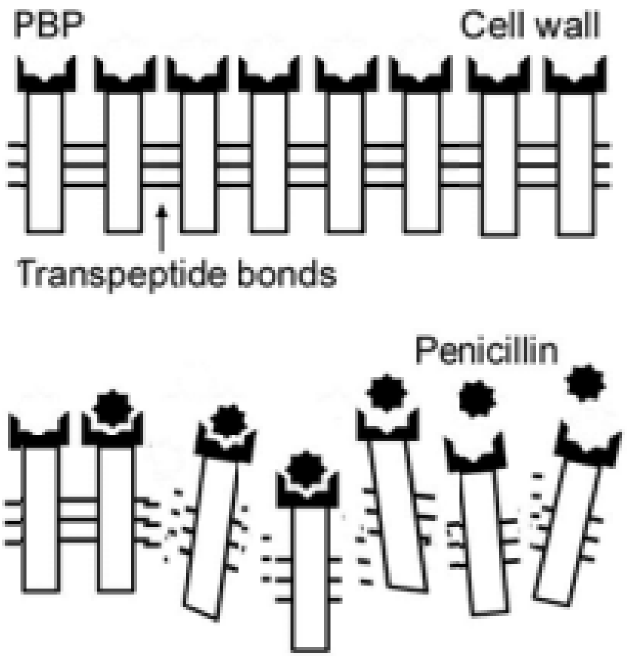
Antibacterial spectrum:

- *Gram-positive cocci,* e.g. streptococci, pneumococci and staphylococci.
- *Gram-negative cocci:* gonococci and meningococci.
- *Gram-positive bacilli:* anthrax bacillus, C. diphtheria and clostridia.
- *Gram-negative bacilli:* shigella, salmonella, pseudomonas…etc.
- *Spirochetes:* Treponema pallidum.
- *Actinomyces.*

University of Alkafeel

Mechanism of action:

- Penicillins bind to **penicillin binding proteins** (PBPs) on the bacterial cell wall and inhibit transpeptidation reaction essential for **bacterial cell wall synthesis.**
- The next step is activation of intracellular autolytic enzymes (**autolysins**) leading to cell rupture. *Thus, the antibacterial effect of penicillin is the result of both inhibition of cell wall synthesis and destruction of existing cell wall by autolysins.*



- Penicillins are **bactericidal** especially for **Gram-positive** bacteria which have thick cell walls.
- The major cause of resistance is the production of **β-lactamase** *(penicillinase)*.

Pharmacokinetics:

- Absorption of oral penicillins is decreased by food. They must be administered 1hr before or 2 hr after meals.
- Route of administration of a B-lactam antibiotic is determined by the stability of the drug to gastric acid and by the severity of the infection.
  - Oral in moderate infection and acid stable preparations e.g. penicillin V.
  - Paraentral in severe infection and acid sensitive preparations e.g. penicillin G
- Penicillins can penetrate to CSF and ocular fluid only during meningitis. They cross placental barrier but they are not teratogenic.
- Most penicillins are excreted through organic acid secretory system via the kidney. The renal excretion in proximal tubules could be decreased by co-administration of probenecid which prolongs its duration of action.

Types of penicillin

	Penicillin G	Anti-staph penicillins	Extended spectrum penicillins	Anti-pseudomonal penicillins
		Cloxacillin Flucloxacillin Nafcillin	**Ampicillin Amoxicillin**	**Ticarcillin Azlocillin Pipracillin**
Spectrum	Gram +ve spp. and few Gram –ve *cocci* *T. pallidum*	*Staphylococci*	Gram +ve and Gram –ve spp. (except *Pseudomonas*)	*Pseudomonas* and Gram –ve spp.
β-lactamase susceptibility	Yes	No	Yes	Yes
			They are combined with **clavulanic acid or sulbactam** to cover β-lactamase producing bacteria.	

1. BENZYL PENICILLIN (PENICILLIN G) and RELATED DRUG:

Effectiveness: active against gram-negative and positive cocci, gram-positive bacilli and spirochetes. It is sensitive to penicillinase enzyme.

Classification:

A. Injectable Short-acting preparation e.g. benzyl penicillin (penicillin-G):

- It is the prototype of all penicillin family.
- Primarily covers **gram-positive organisms** (*Pneumococci, streptococci, staphylococci* (non-penicillinase-producing), *Cl. perfringens,C. diphtheria*) with **little gram-negative** coverage (*Gonococci* and *meningococci[but not bacilli]*), making it **ineffective** in urinary tract infection.
- ***Others:*** *Treponema pallidum* (syphilis).
- It has some limitations:
  - It is acid labile and must be given parentrally.
  - It has a short half-life, so frequent injections are required.
  - It is easily inactivated by β-lactamase enzyme.
  - It has narrow spectrum.

B. Injectable Long-acting preparation

i. Penicillin G Procaine:

It is a suspensions of combination of penicillin G with procaine that have longer duration (12-24 hrs) allowing reduced frequency of injections. So it is injected just once every day, intramuscularly.

ii. Benzathine Penicillin: This preparation produces low blood levels lasting from few days to 4 weeks depending on the dose. So it is used in chemoprophylaxis and in the treatment of syphilis.

C. Oral Penicillin V: it is an acid stable form of penicillin G (given orally).

2.PENICILLINASE OR β-LACTAMASERESISTANT PENICILLINS (ANTI-STAPH PENICILLINS)

[Methicillin, cloxacillin, dicloxacillin, flucloxacillin, and **nafcillin]**

- They are effective against β-lactamase producing*Staphylococci*.
- They have low or **no activity** against **other gram-positive and gram-negative organisms**.
- The use of these agents is now declining due to increased incidence of methicillin-resistant *S. aureus* (MRSA).

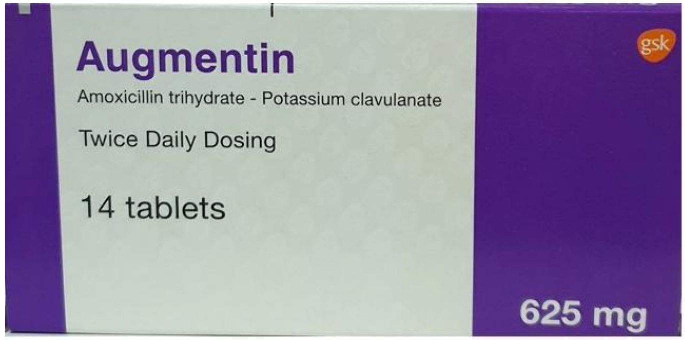
3. Extended spectrum penicillins

[Ampicillin and **amoxicillin]**

- They maintain gram-positive coverage with extended coverage of more **gram-negative bacteria** e.g. *Salmonella*, H. influenza, Proteus, and Shigella (but not *Pseudomonas*).
- They can be given orally (acid resistant) and by injection (IM and IV).
- They are inactivated by β-lactamase, so they are usually combined with **clavulanic acid** or **sulbactam** (β-lactamase inhibitors) to cover β-lactamase producing bacteria. **[Amoxicillin/clavulanic acid** (co-amoxiclav) - **Ampicillin/sulbactam** (Unasyn)].
- **Ampicillin** is concentrated in bile so it is effective in gall bladder disease and typhoid carrier.

Common organisms capable of producing penicillinase:

Staphylococcus aureus,
Escherichia coli,
H. influenzae
Pseudomonas aeruginosa
Neisseria gonorrhoeae
Proteus
***Bacteroides*species.**



- **Ampicillin**is more active on shigella and H. influenza.
- **Pro-drugs of ampicillin** e.g. pivampicillin and talampicillin are esters of ampicillin which themselves are microbiologically inactive but after oral administration they are de-esterified in the gut mucosa or liver to release ampicillin to the systemic circulation. They are better absorbed from the gut than ampicillin itself, so gives higher levels in blood and tissues and they have no effect on the gut flora and less G.I. upset.
- **Amoxicillin** has better absorption and tissue penetration than ampicillin. So, it has no effect on the gut flora and less G.I. upset.
- **Amoxicillin** is more active against Salmonella and Streptococcal fecalis.
- **Amoxicillin** can penetrate mucoid and purulant sputum, so it is useful in chronic bronchitis.

4.ANTIPSEUDOMONALPENICILLINS

[Ticarcillin, Azlocillin, and **Piperacillin]**

- They are broad-spectrum, but should only be used for pseudomonas infection and ampicillin resistant proteus. Also they have activity against anaerobic gram negative bacteria e.g. Bacteroids fragilis (a common pathogen in intra-abdominal sepsis)
- They have synergistic effect when they are used with aminoglycosides.
- They are inactivated by β-lactamase so they are combined with **clavulanic acid forming Timentin** to cover β-lactamase producing bacteria.

Therapeutic Uses of Penicillins

A. Treatment: Penicillins may be used in the treatment of:

- Streptococcal infections, e.g. wound sepsis, acute throat infections, subacute bacterial endocarditis,.. etc.
- Staphylococcal infections of skin, mucous membrane and bone.
- Pneumococcal infections e.g. pneumonia.
- Syphilis (5 million units single dose) and gonorrhoea.
- Meningococcal infections: Penicillin diffuses into the CSF only when the meninges are acutely inflamed.
- Typhoid and paratyphoid fevers: ampicillin and amoxycillin.
- Pseudomonas infection: (ticarcillin or piperacillin)
- Actinomycosis, Anthrax and H. influenza infections.
- Diphtheria, tetanus and gas gangrene. (Penicillin may be used together with the specific antitoxins).

B. Prophylaxis: Penicillins may be used prophylactically in the following conditions:

- To prevent recurrence of rheumatic fever: Benzathine penicillin, 1.2 million units IM once a month.
- To prevent subacute bacterial endocarditis due to bacteraemia resulting from operative procedures such as dental extraction, tonsillectomy…etc. in patients with congenital or acquired valvular disease or immunocompromised patient.

- **Hypersensitivity reactions:**
  - It occurs in ~10% of patients receiving penicillin.
  - It occurs with all types of penicillin. Allergy is not due to penicillin itself but to degradation product common to all penicillin
  - It is more common after parenteral than oral administration.
  - All types of reactions, from simple rash to **acute anaphylaxis and angioedema**, can occur within 2 minutes (**early or type I** hypersensitivity) or up to 12 days (**delayed or type II**) after administration.

Adverse Effects of Penicillins

- True anaphylaxis is rare (1:10’000 of cases).
- Penicillin allergy is unpredictable i.e. an individual who tolerated penicillin in the past may develop allergy later on and *vice versa*.

Prevention:

- Never give penicillin if there is history of penicillin allergy.
- Test for hypersensitivity

Management of anaphylactic shock: see CVS.

Drug interactions:

- **Bacteriostatic drugs** (e.g. tetracycline, chloroamphenicol, erythromycin) interfere with the action of penicillin because penicillin acts by inhibiting cell wall synthesis (i.e. kills rapidly multiplying bacteria) while bacterio**static** drugs arrest the organism and prevents its multiplication.
- **Antipseudomonal penicillins** (acidic drugs, —ve charged) form complex with **amin-oglycosides** (basic drugs, +ve charged) if they are mixed in the same infusion fluid.